

# Probabilistic Graphical Models

## CSE 5830

### Introduction

What are PGMs and  
probabilistic graphical models?

# What are Probabilistic Graphical Models (PGMs)

- Marriage between Computer Science (graph theory, machine learning) and Statistics (probability theory).
- A powerful framework which can be used to learn models with dependency
- Two problems: complexity and uncertainty
  - Multivariate statistics plus structure

# Why PGMs?

- Real-world problems involve uncertainty and complex dependencies (e.g., medical diagnosis, speech recognition).
- Traditional models often assume independence → too simplistic.
- Need a structured way to represent joint distributions compactly.

Example: Diagnosing disease from symptoms — dependencies exist among symptoms and diseases.

# Reasoning or inference

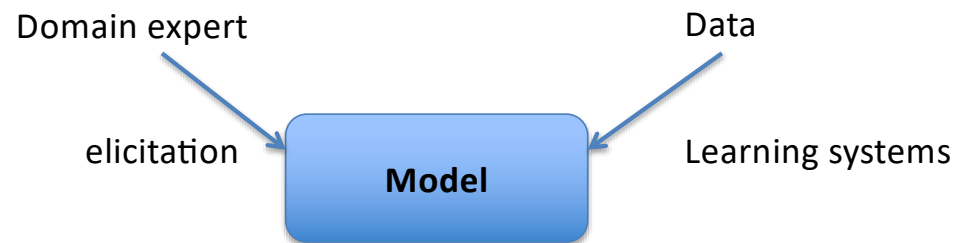
- Take the available information and reach conclusions about what is true and how to act
  - Doctors
  - Speech recognition systems
  - Robots
- We want a general framework that can be used to allow a computer system to make an inference from very complex systems
- We want to take advantage of the structure, more specifically, independences and conditional independences that hold among random variables

# Why Use Probabilistic Graphical Models?

- **Compact representation** of complex joint distributions using conditional independencies.
- **Interpretability:** graph structure reveals how variables depend on each other.
- **Inference:** enables reasoning about hidden variables given observed data.
- **Integration of domain knowledge** with data-driven learning.
- **Modularity:** can model complex systems hierarchically.

# Probabilistic Graphical Models

- Declarative representation
  - Construct a model of the system about which we would like to reason
  - Independent of the algorithms used for reasoning
    - Separation of knowledge and reasoning



# Probabilistic Graphical Models

- Model complex systems with a significant amount of uncertainty
- True state of the system is not clear
  - Observations are not complete
  - Observations are noisy
  - Nothing is deterministic

To reason not just about what is possible, but also about what is probable

# Probabilistic Graphical Models

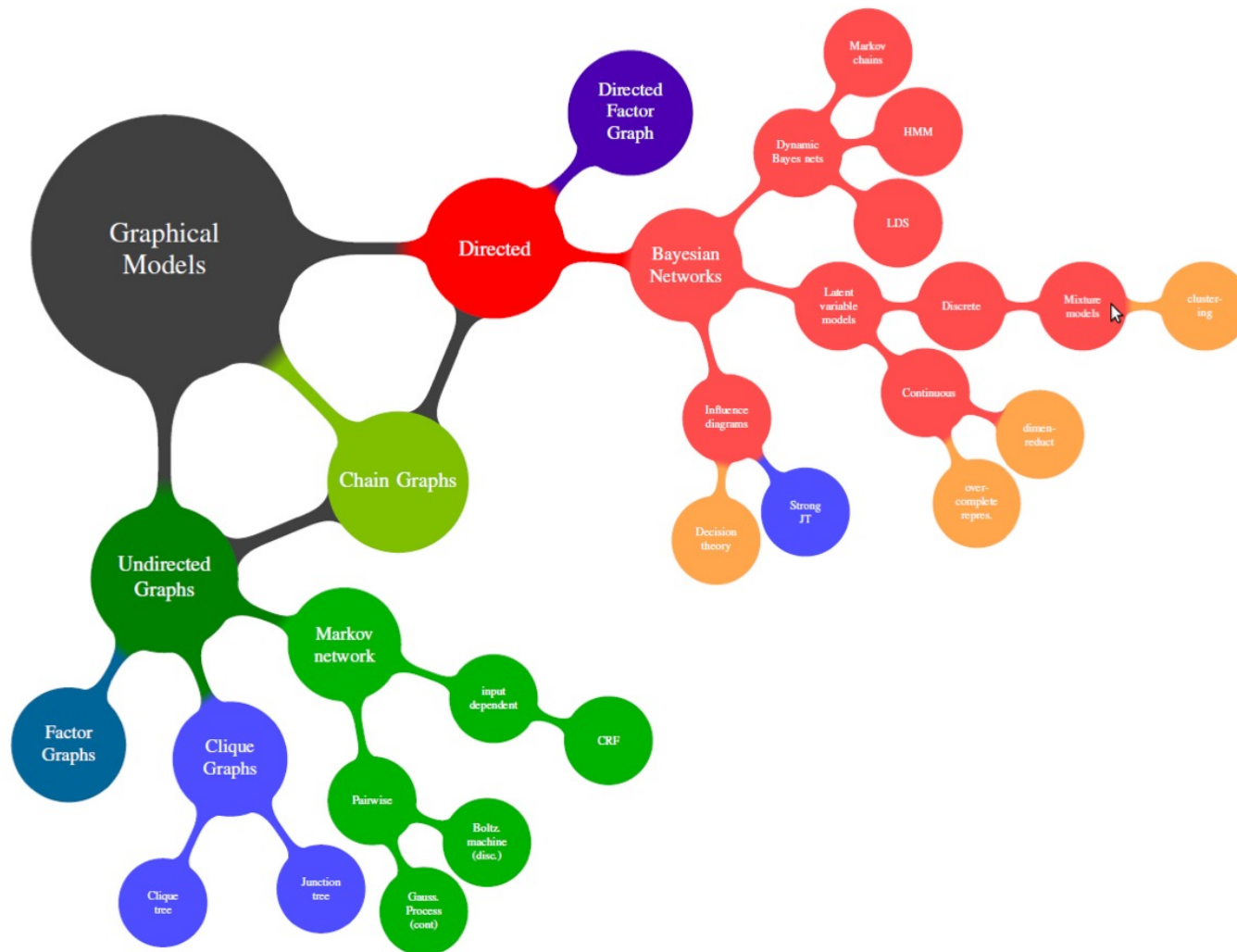
- Probabilistic models of complex systems
- Complex systems are characterized by the presence of multiple interrelated aspects
- Graph can be used to represent complex systems with large number of variables
  - Random variables

# Joint distribution

- To reason probabilistically about the values of some random variables, given observation about some others
  - Random variables  $x_1, \dots, x_n$
  - Joint distribution  $P(x_1, \dots, x_n)$
- PGMs use a graph-based representation as the basis for compactly encoding complex distribution over a high-dimensional space.

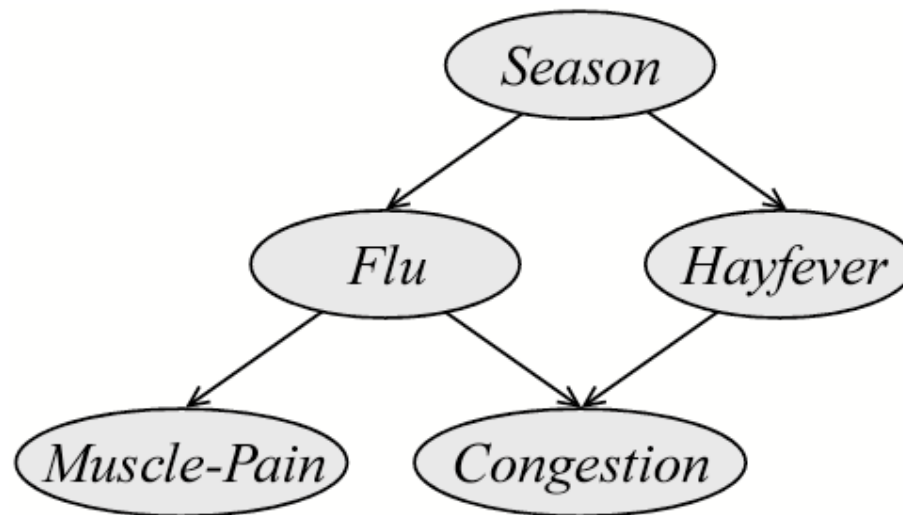
# Graph-based representation

- Graph
  - nodes and edges
- Nodes represent variables
- Edges represent direct probabilistic interaction between two nodes



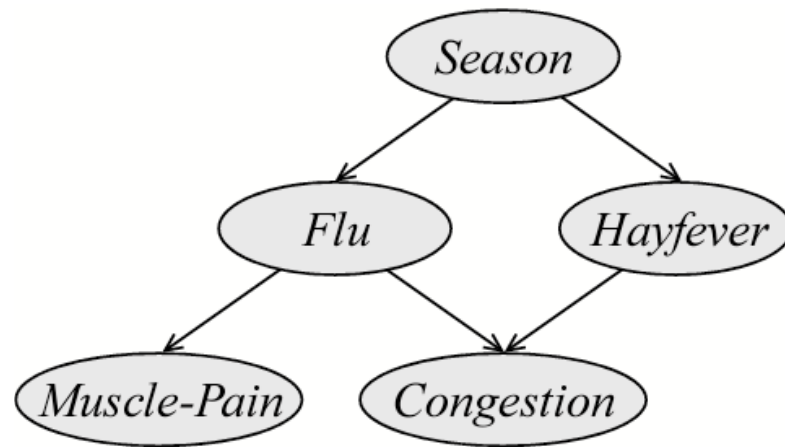
# Bayesian network

Graph Representation



# Bayesian network

$$\begin{aligned} &(F \perp H \mid S) \\ &(C \perp S \mid F, H) \\ &(M \perp H, C \mid F) \\ &(M \perp C \mid F) \end{aligned}$$

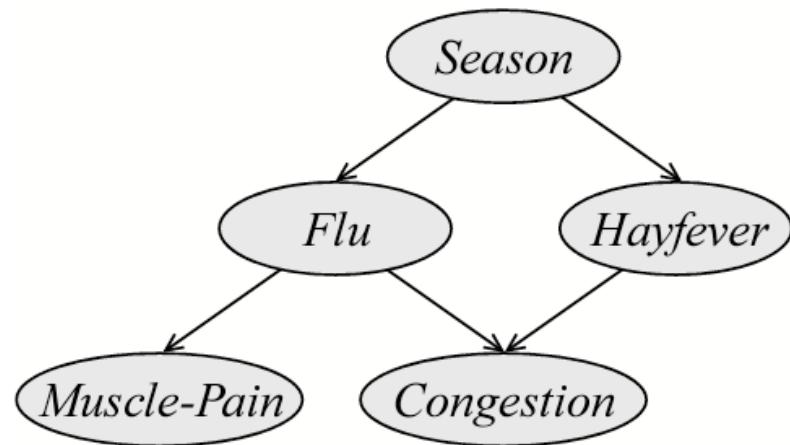


# Bayesian network

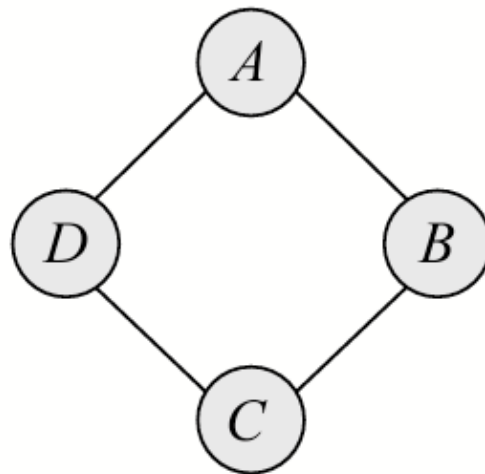
- Represent the full joint distribution over the variables more compactly using a smaller number of parameters.
- Take advantage of conditional and marginal independences among random variables

$$\begin{aligned} &(F \perp H \mid S) \\ &(C \perp S \mid F, H) \\ &(M \perp H, C \mid F) \\ &(M \perp C \mid F) \end{aligned}$$

$$\begin{aligned} P(S, F, H, C, M) &= P(S)P(F \mid S) \\ &\quad P(H \mid S)P(C \mid F, H)P(M \mid F) \end{aligned}$$

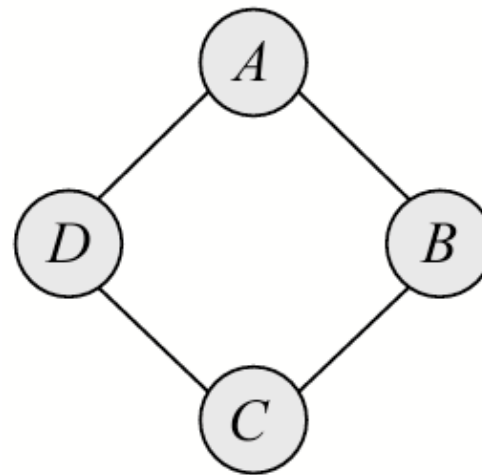


# Markov network



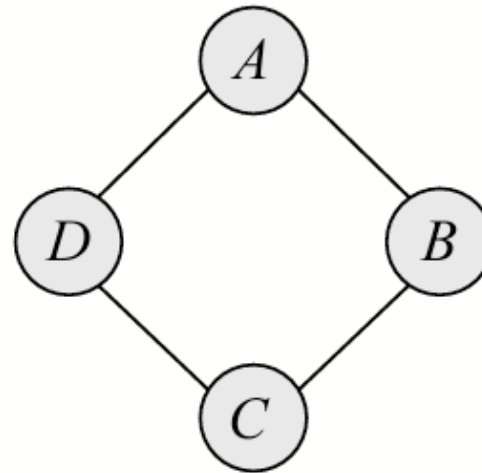
# Markov network

$$\begin{aligned} (A \perp C \mid B, D) \\ (B \perp D \mid A, C) \end{aligned}$$



# Markov network

$$\begin{aligned} (A \perp C \mid B, D) \\ (B \perp D \mid A, C) \end{aligned}$$



$$P(A, B, C, D) = \frac{1}{Z} \phi_1(A, B) \phi_2(B, C) \phi_3(C, D) \phi_4(A, D)$$

# Probabilistic Graphical Models

- Framework for obtaining, representing, querying large probability distributions.
- A beautiful formalism that generalizes many ideas from Computer Science and from Statistics.

# The fundamental questions

- Representation
  - How to capture/model uncertainties in possible worlds?
  - How to encode our domain knowledge/assumptions/constraints?
- Inference
  - How do I answer questions/queries according to my model and/or based given data?

$$\text{e.g.: } P(X_i \mid \mathbf{D})$$

- Learning
  - What model is "right" for my data?

$$\text{e.g.: } M = \underset{M \in \mathcal{M}}{\operatorname{argmax}} F(\mathbf{D}; M)$$

$$M \in \mathcal{M}$$

# Representation, inference, learning

Probabilistic graphical models:

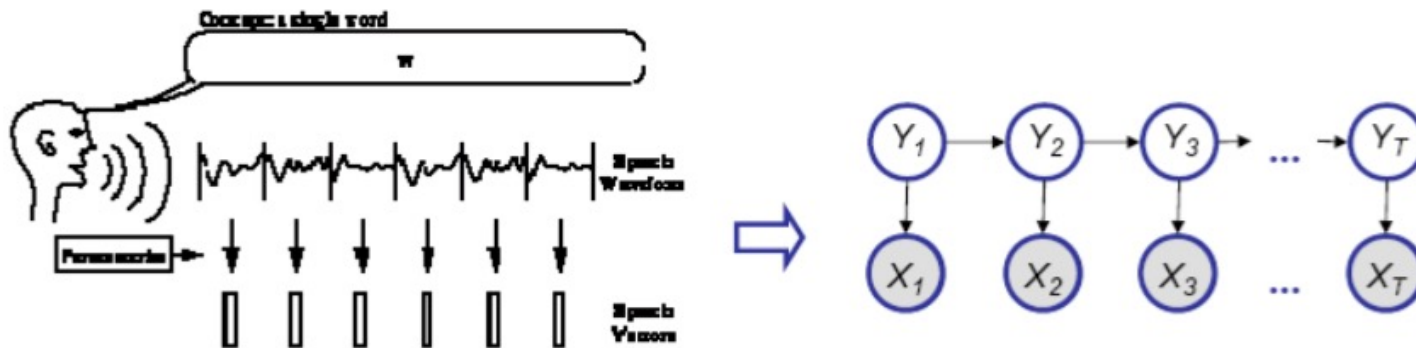
The use of structure to allow a compact representation, effective reasoning, and feasible learning of general propose, factored, probabilistic models.

# PGMs have many applications

- Speech recognition (hidden Markov models)
- Tracking and robot localization (Kalman filters)
- Evolutionary biology (Bayesian networks)
- Modeling sensor data (undirected GMs)
- Image segmentation (hierarchical BNs)
- Natural language processing (probabilistic grammars)
- Structured data like social networks and linked documents (probabilistic relational models)
- Decoding messages over noisy channels (belief propagation)
- Biological network reconstruction
- ...

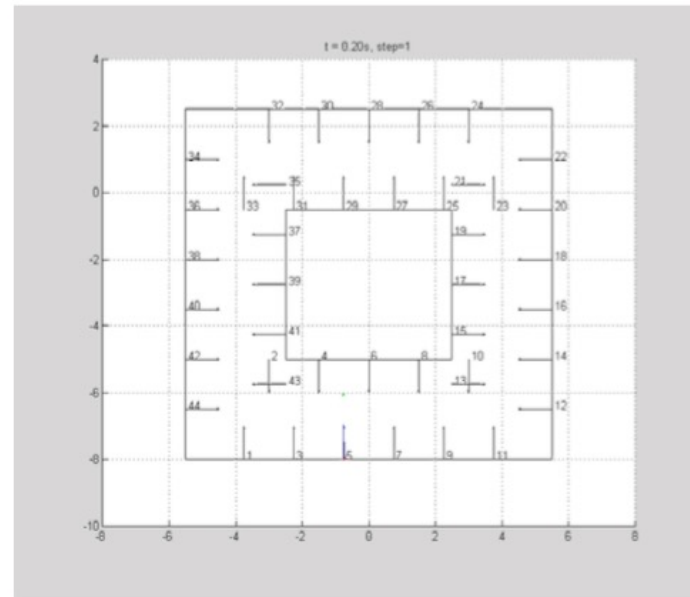
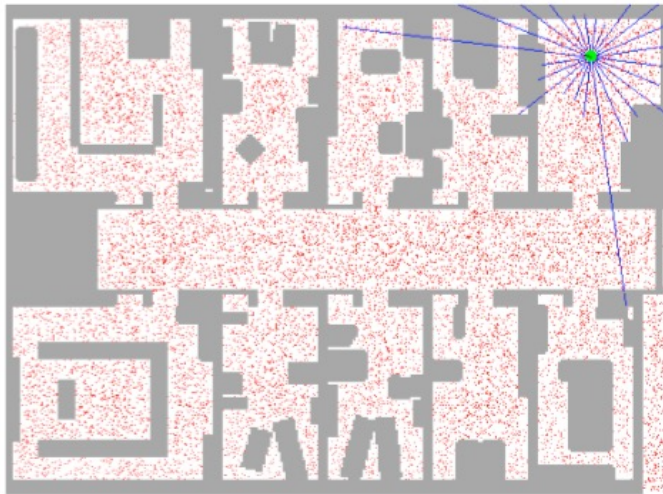
# PGMs have many applications

Speech recognition (hidden Markov models)



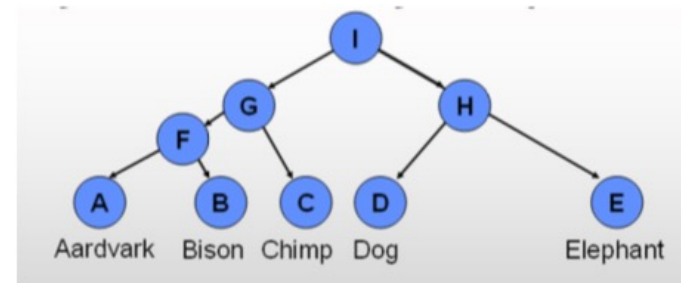
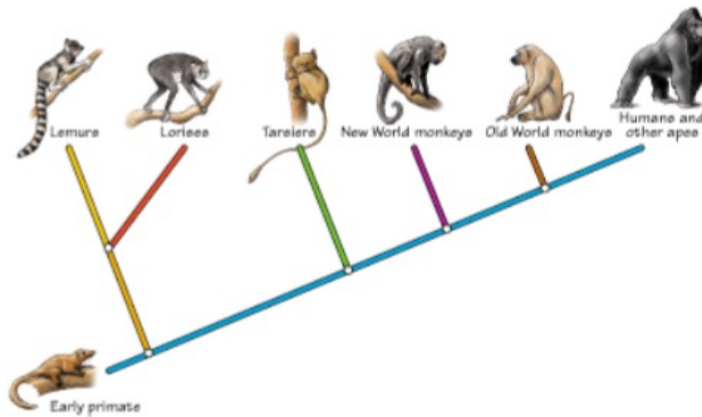
# PGMs have many applications

Tracking and robot localization (Kalman filters)



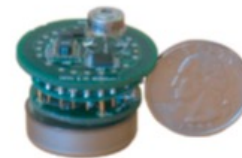
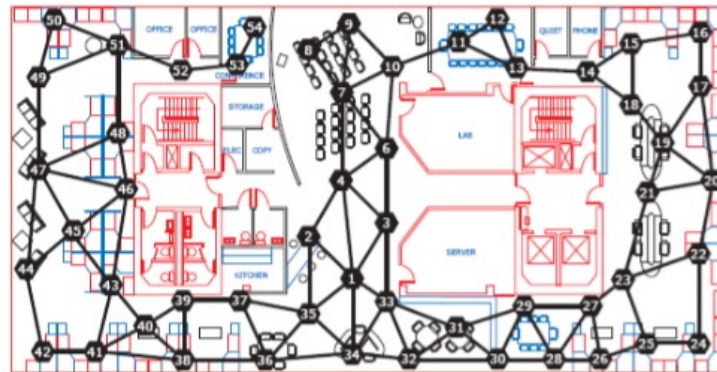
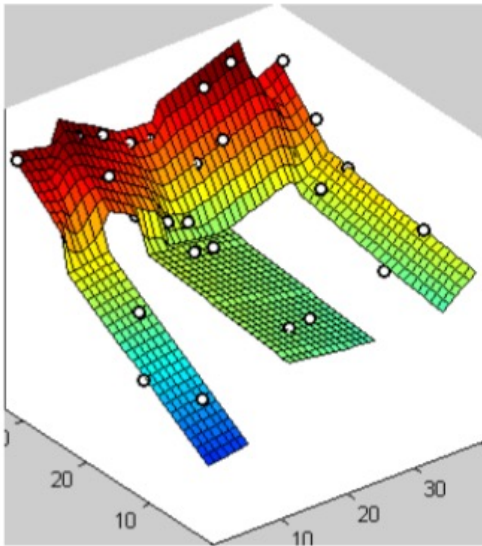
# PGMs have many applications

## Evolutionary biology (Bayesian networks)



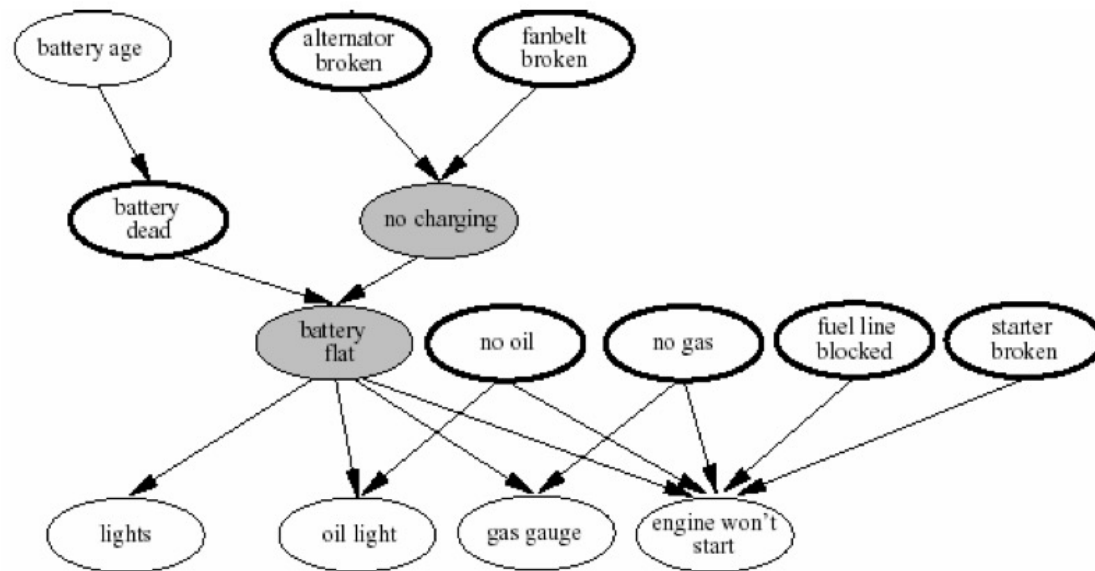
# PGMs have many applications

Modeling sensor data (undirected GMs)



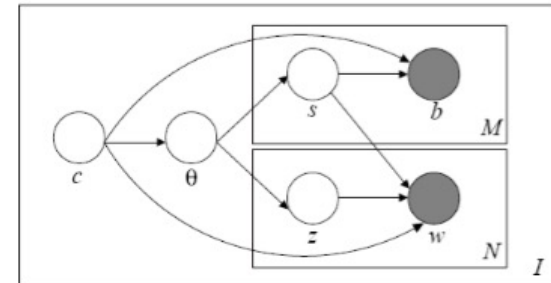
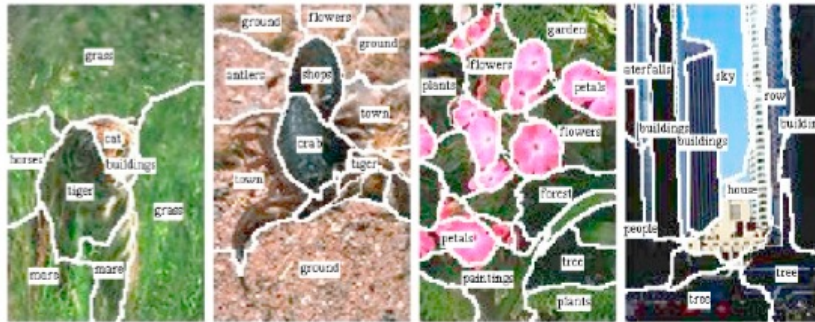
# PGMs have many applications

## Diagnostic of car engine (Bayesian Network)



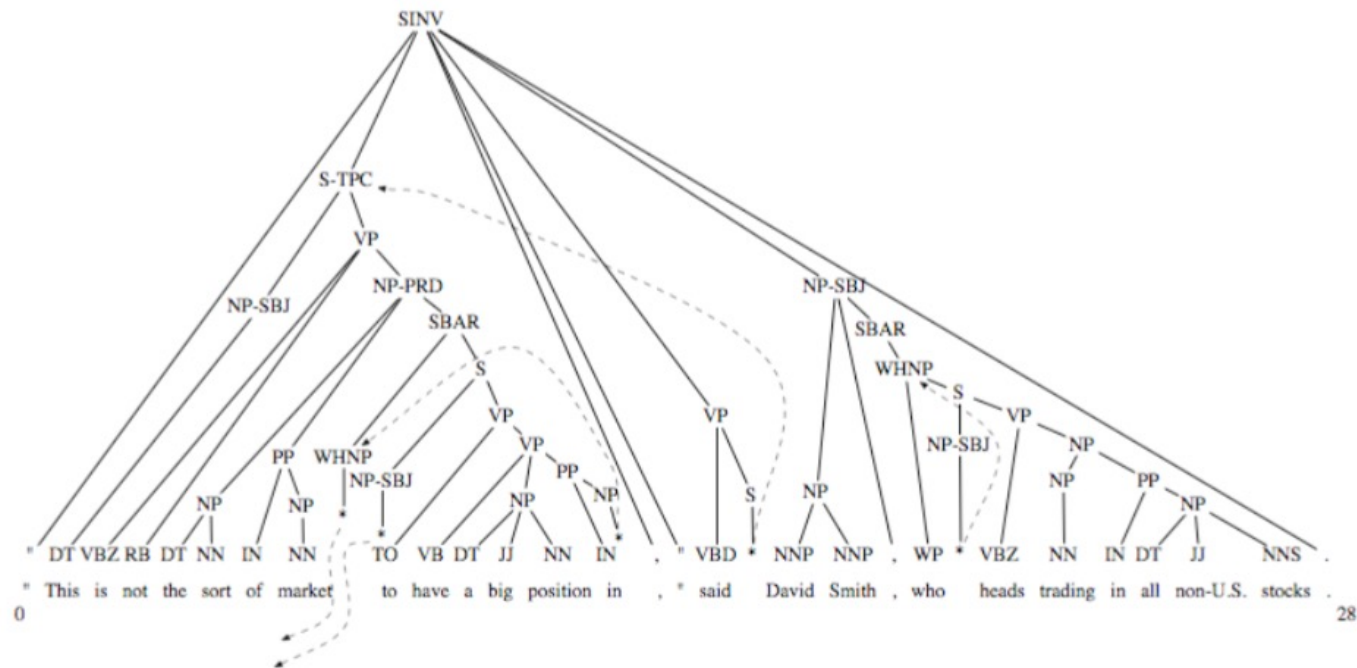
# PGMs have many applications

## Image segmentation (hierarchical BNs)



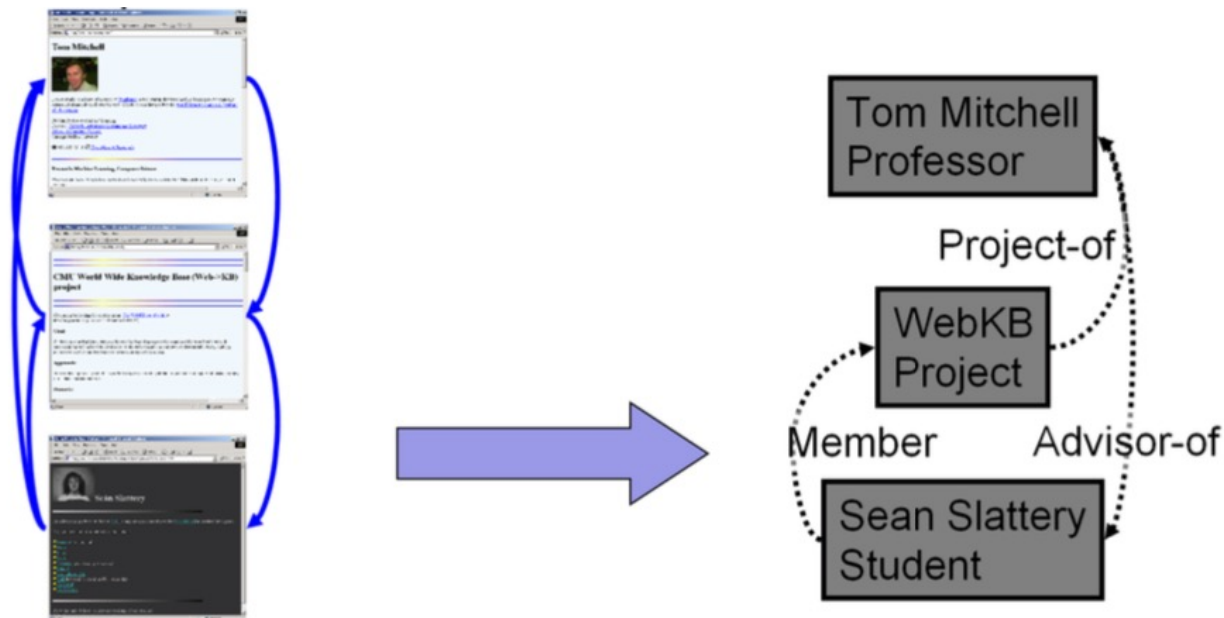
# PGMs have many applications

Natural language processing (probabilistic grammars)



# PGMs have many applications

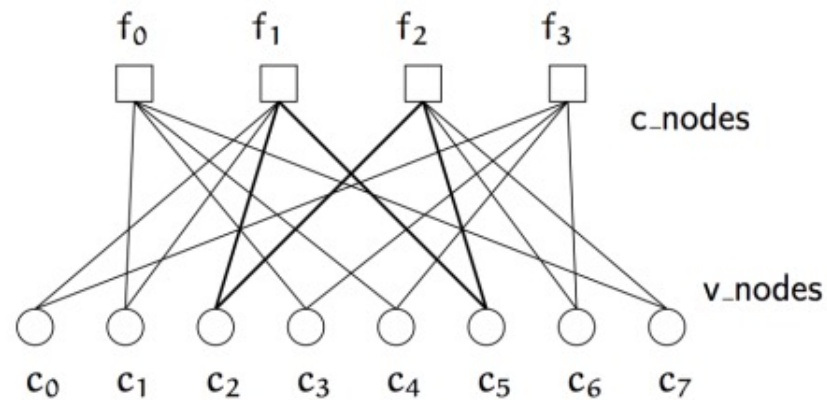
Structured data like social networks and linked documents  
(probabilistic relational models)



# PGMs have many applications

Decoding messages over noisy channels (belief propagation)

$$\mathbf{H} = \begin{bmatrix} 0 & 1 & 0 & 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \end{bmatrix}$$

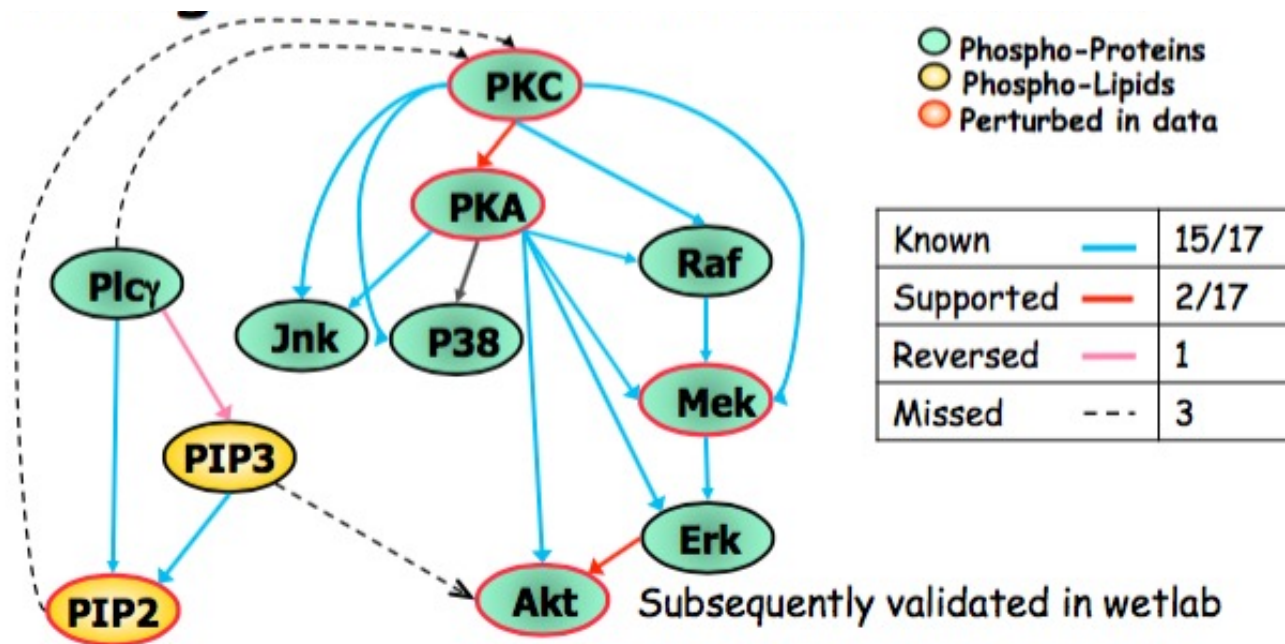


## Prediction and Analysis of COVID-19 with Probabilistic Graphical Models



# PGMs have many applications

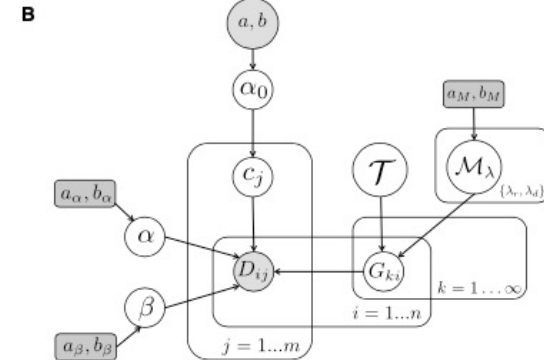
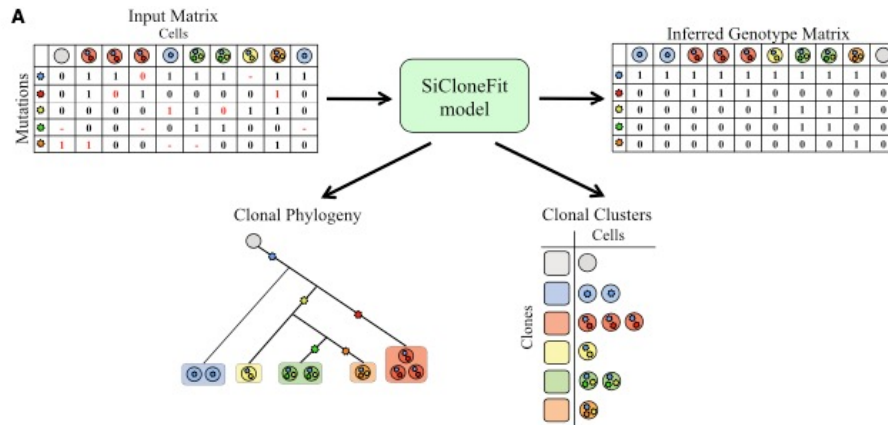
## Biological network reconstruction



# PGMs have many applications

## Method

**SiCloneFit: Bayesian inference of population structure, genotype, and phylogeny of tumor clones from single-cell genome sequencing data**



**C**

$$\alpha | a_\alpha, b_\alpha \sim \text{Beta}(a_\alpha, b_\alpha)$$

$$\beta | a_\beta, b_\beta \sim \text{Beta}(a_\beta, b_\beta)$$

$$\alpha_0 \sim \text{Gamma}(a, b)$$

$$c_j | \alpha_0 \sim \text{CRP}(\alpha_0)$$

$$\mathcal{T} \sim T_{\text{prior}}(K)$$

$$\mathcal{M}_\lambda | a_M, b_M \sim \text{Beta}(a_M, b_M)$$

$$G_{ki} | \mathcal{T}, \mathcal{M}_\lambda \sim F(G_{ki} | \mathcal{T}, \mathcal{M}_\lambda)$$

$$D_{ij} \sim E(D_{ij} | G_{c,i}, \alpha, \beta)$$

# The Reality Check: The Curse of High Dimensions

- The Core Conflict — Where Classical PGMs Break
- Classical PGMs: Excellent for structured, tabular, or domain-expert features  $(X_1, X_2, \dots, X_n)$ .
- The Bottleneck: What happens when  $X$  is a 4K image? Or an entire genome sequence?
  - Conditional Probability Tables scale exponentially with continuous or high-dimensional variables.
  - Junction Trees quickly become unmanageable in the real world (Treewidth restrictions) .

# Can Deep Learning Help?

- Neural networks provide highly expressive function approximators for conditional dependencies.
- Enables modeling of complex, high-dimensional, nonlinear relationships.
- Learns features automatically from raw data (images, text, sequences).

Deep neural networks are seen as deterministic. Can it be framed probabilistically?

# Why Probabilistic Deep Learning

- Pure deep learning lacks uncertainty quantification and probabilistic reasoning.
- Probabilistic deep learning combines the strengths of both worlds:
  - Expressive representations (deep learning)
  - Structured uncertainty modeling and inference (PGM)
- Enables robust, interpretable, and data-efficient AI systems.

# Deep Learning as a PGM

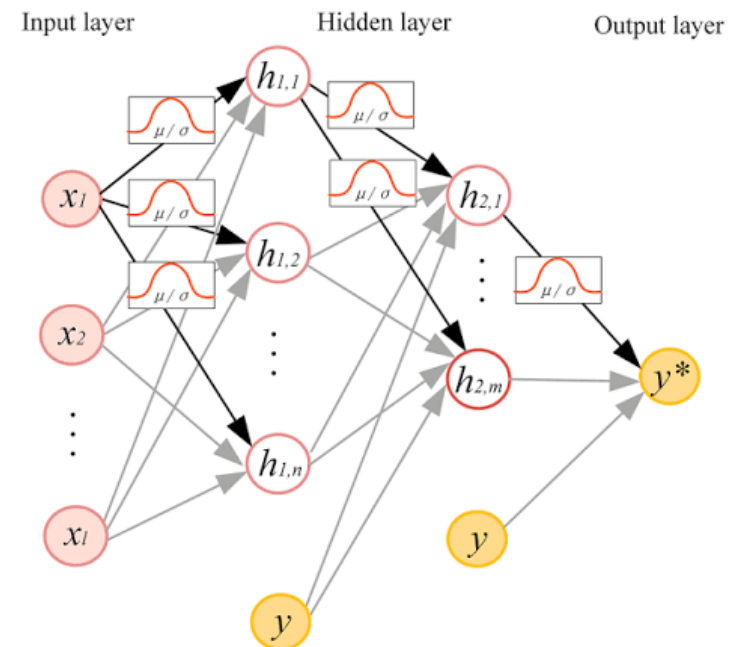
- Any feedforward neural network can be viewed as a directed graphical model:
  - Input nodes  $\rightarrow$  hidden nodes  $\rightarrow$  output nodes.
  - Each layer defines a conditional distribution:  $p(h_l | h_{l-1}; \theta_l)$ .
- In deterministic networks:  $h_l = f_\theta(h_{l-1})$
- In probabilistic form:  $h_l = p_\theta(h_{l-1})$ .
- Thus, deep learning can be viewed as a hierarchical latent variable model
  - a deep PGM.

# Probabilistic deep neural networks

- Probabilistic deep learning = Deep neural networks + Probabilistic reasoning.
- It combines the representation power of deep learning with the rigor of probability, allowing models not only to predict but also to know how sure they are.
- Both describe complex systems in terms of random variables and their probabilistic dependencies.
  - PGMs (e.g., Bayesian networks, Markov random fields) represent those dependencies explicitly using a graph structure (nodes = variables, edges = conditional dependencies).
  - PDL represents them implicitly through the parameters and structure of deep neural networks, often learned end-to-end.

# Deep & Modern Probabilistic Graphical Models

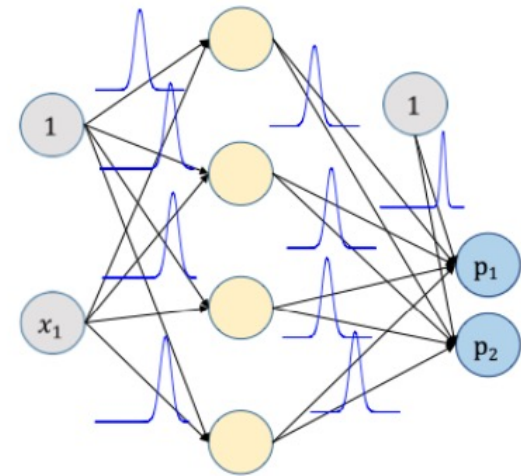
- Bridging Structural inference with Deep Neural Network Scale
- Exploiting everything we know about probability graphs and scale them up using the raw representation power of Deep Learning



Bayesian Deep learning

# Bayesian Neural Network

- Moving from Deterministic Weights to Parameter Random Variables
- Each weight is replaced by a distribution.
  - can be complicated.
- Posterior distributions of the weights are approximated by a simple distribution, called *variational distribution*
- Often Gaussians are used as parametric distributions
  - learn the two parameters of weight distribution: the mean and the variance.



# Deep Generative Architectures

Generative models, defining a distribution  $P(X|Z)$  (density estimation)

- Variational Autoencoders (VAEs)
  - Shallow directed graphs ( $z \rightarrow x$ )
  - compress data into a low-dimensional continuous latent
- Markov Diffusion
  - Deep sequential directed graphs ( $x_T \rightarrow \dots \rightarrow x_1 \rightarrow x_0$ )
  - break generation into hundreds of tiny, discrete or continuous denoising steps
  - forward/reverse stochastic process.

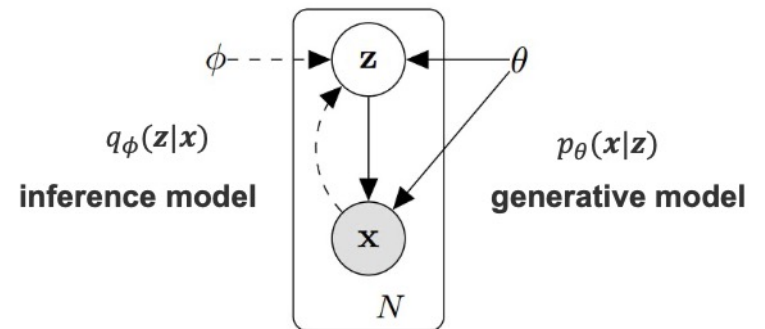
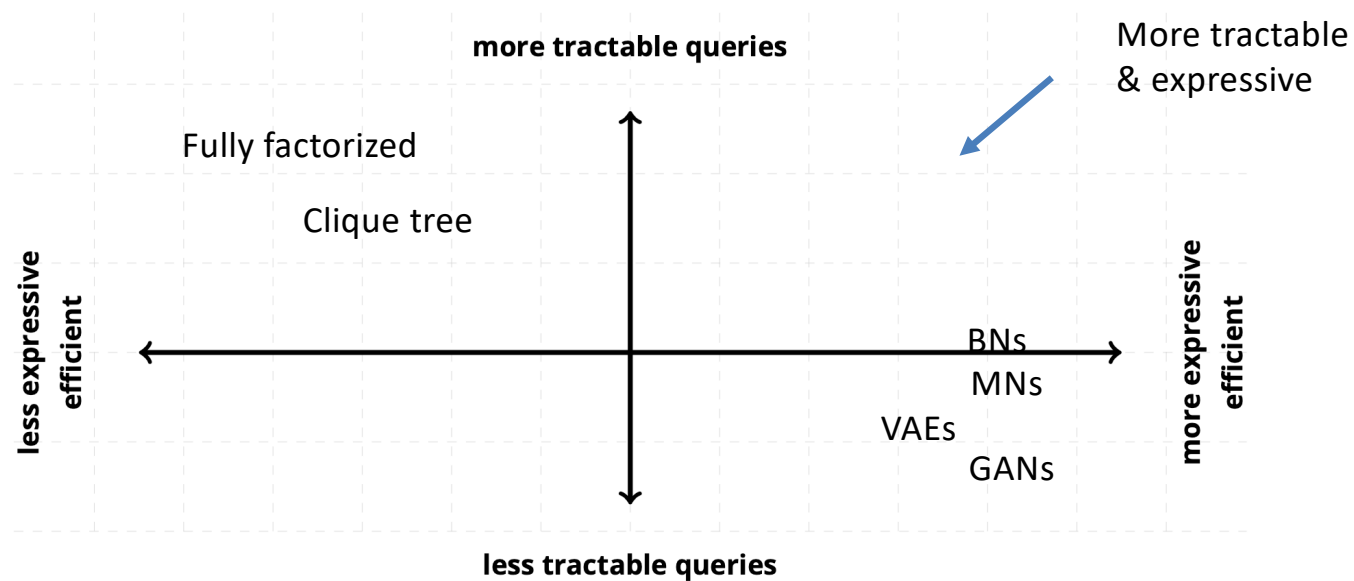


Figure courtesy: Kingma & Welling, 2014

# Generative Adversarial Networks (GAN)

- GANs are deep generative models capable of learning complex data distributions.
- They bridge probabilistic modeling and game theory.
- Provide a framework to generate realistic data (images, text, etc.) without explicitly defining a likelihood.
  - GANs learn implicit distributions via adversarial training.
- Key idea: Learn to generate samples that are indistinguishable from real data.

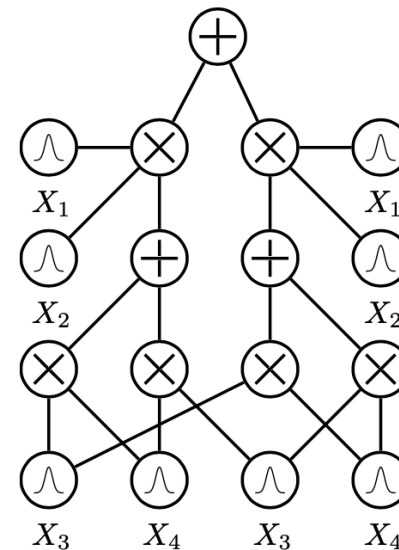
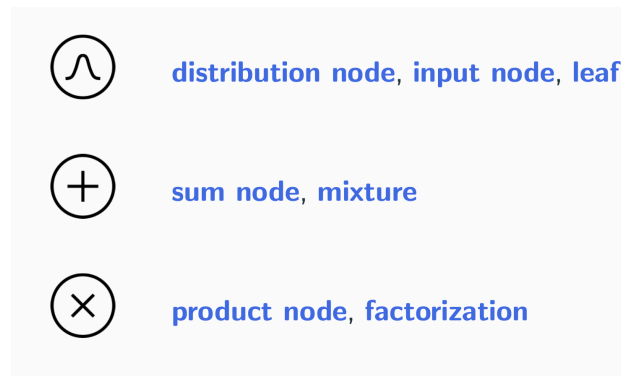
# Tractable vs expressive



- Expressive models are not very tractable
- and tractable ones are not very expressive

# Probabilistic Circuits (PCs):

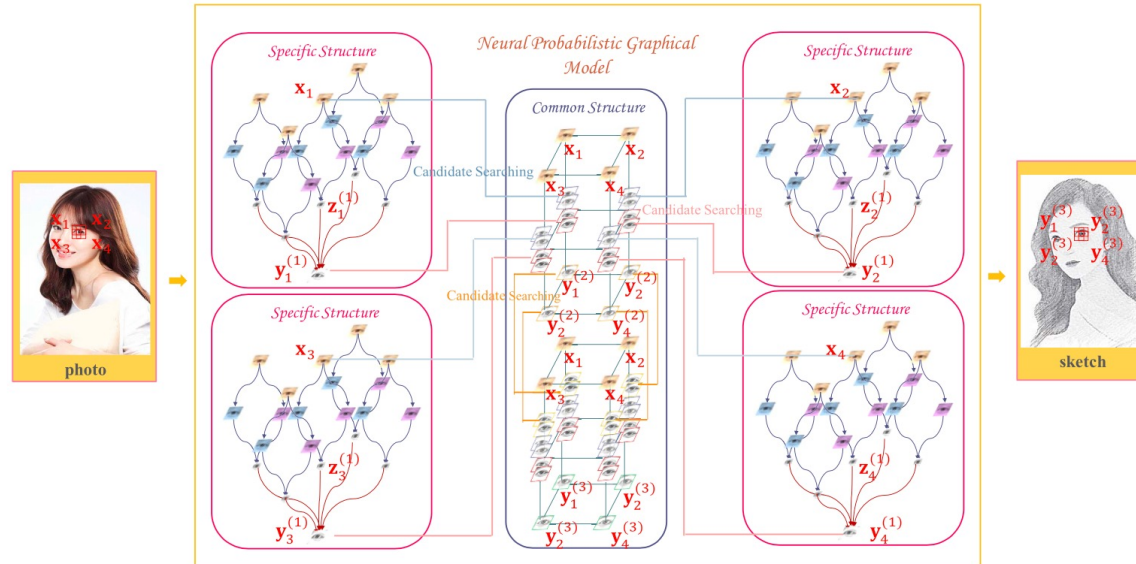
- Special type of neural network
- represent a density  $p(X)$  for given random variables  $X$
- basic mode of operation: “sample  $x$  in,  $p(x)$  out”



# PDLs have many applications

IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, VOL. 31, NO. 7, JULY 2020

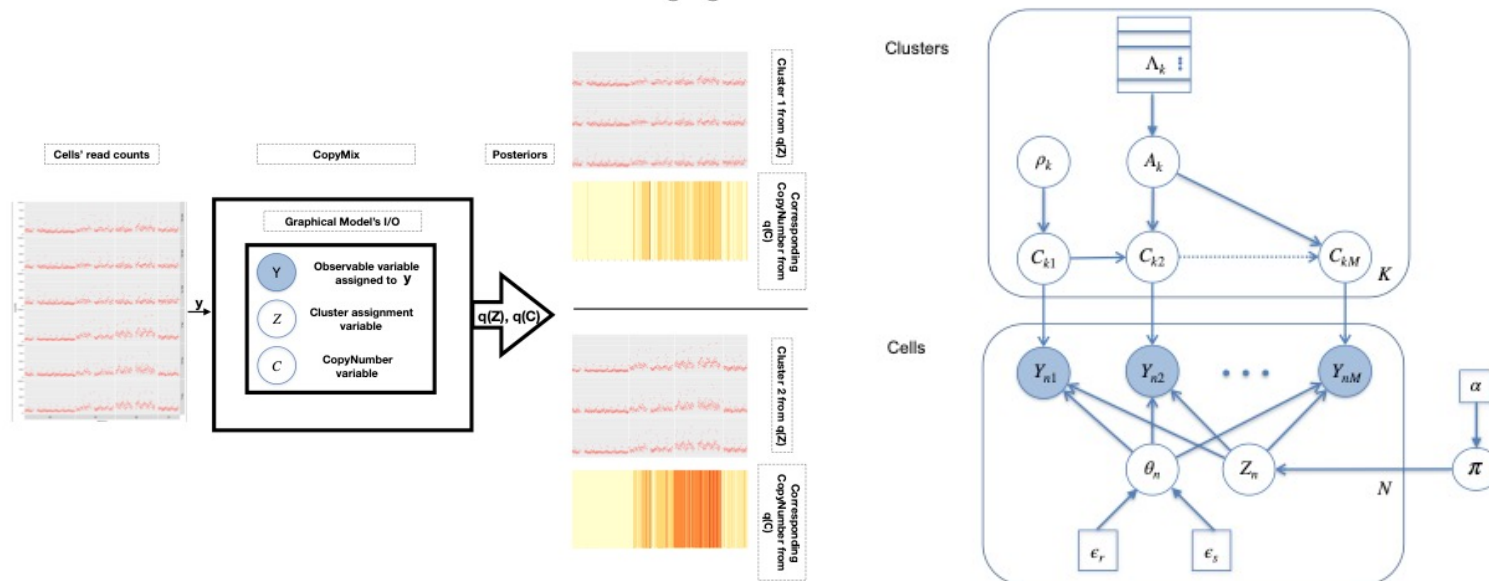
## Neural Probabilistic Graphical Model for Face Sketch Synthesis



# PDLs have many applications

## CopyMix: Mixture Model Based Single-Cell Clustering and Copy Number Profiling using Variational Inference

Negar Safinianaini, Camila P. E. de Souza<sup>2</sup>, Andrew Roth<sup>4,5</sup>, Hazal Koptagel<sup>1,3</sup>, Hosein Toosi<sup>1,3</sup>, Jens Lagergren<sup>1,3</sup>



# PDLs have many applications

2025 5th International Conference on Artificial Intelligence and Industrial Technology Applications (AIITA)

## Addressing Class Imbalance with Probabilistic Graphical Models and Variational Inference

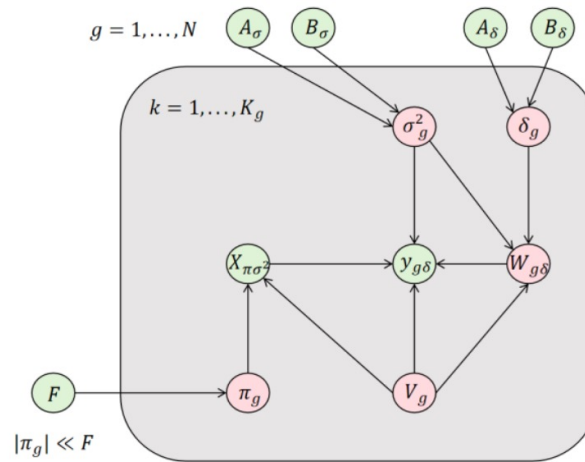


Figure 1. The architecture of the probabilistic graphical model

# PDLs have many applications

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## PGM-Explainer: Probabilistic Graphical Model Explanations for Graph Neural Networks

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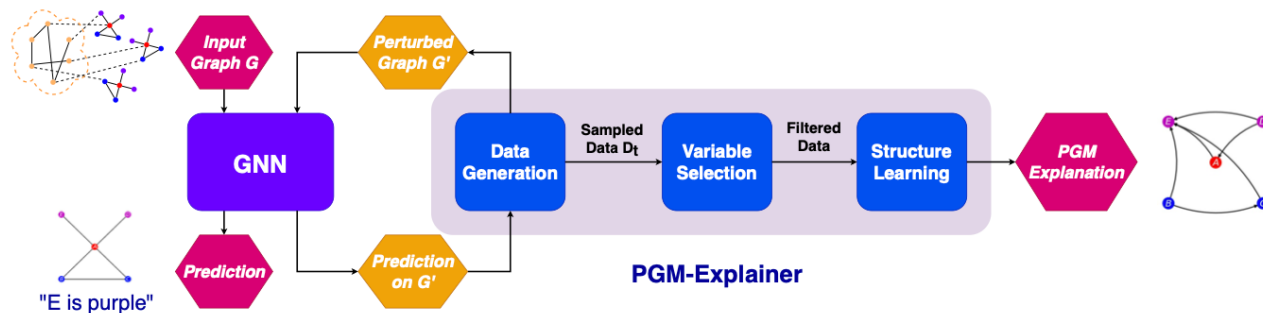
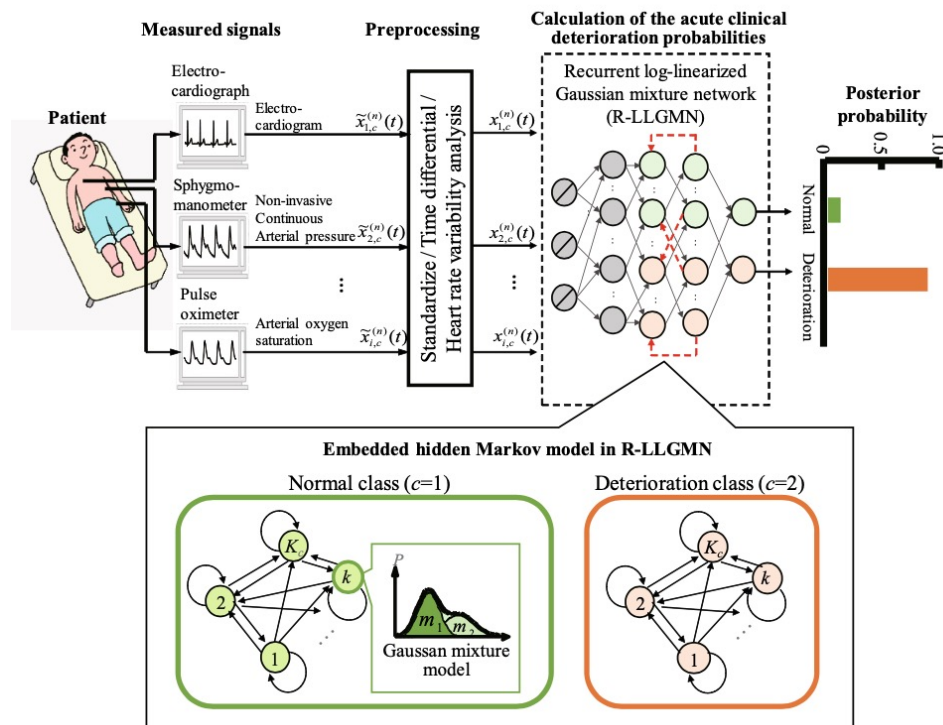


Figure 2: The architecture of PGM-Explainer. Given input graph  $G$  and a prediction to be explained, PGM-Explainer generates perturbed graphs and records GNN's predictions on those graphs in the data generation step. The variable selection step eliminates unimportant explained features in this data and forwards the filtered data. Finally, the PGM is generated in the structure learning step.

# PDLs have many applications

Recurrent probabilistic neural network-based short-term prediction for acute hypotension and ventricular fibrillation



# PGMs have many applications

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## What Uncertainties Do We Need in Bayesian Deep Learning for Computer Vision?

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